



Department of Information Technology
Academic Year 2023 - 2024 (Odd Semester)

Degree, Semester & Branch: III Semester B.Tech. IT

Course Code & Title: CS3352 & Foundations of Data science

Name of the Faculty member (s): Dr.V. Anusuya, ASCP/IT

Innovative Practice Description

- **Unit / Topic:** Unit I / Introduction
- **Course Outcome:** CO1
- **Topic Learning Outcome:** TLO2
- **Activity Chosen:** Reflection
- **Justification:**

In unit I, Basic Statistical descriptions of Data is an important topic needed for every data scientist because the statistics is fundamental for data science. This activity helps the students to know the statistical terminology.

- **Time Allotted for the Activity:** 5 Minutes
- **Details of the Implementation:**
 - I explained the topic Basic statistical Description of data in the previous days.
 - The time allotted for the topic is 5 Minutes. The students during the activity as shown in Figure 1,2
 - The students were asked to study the concepts Mean, Median, Mode, Standard Deviation and Variance
 - Next day asked the students write their doubts in a paper.
 - The papers were collected and corrected as shown in Figure 3

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO12	PSO1	PSO2	PSO3
CO1	3	3	2	2	2	3	2

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO3	PO12	PSO1	PSO2	PSO3
	3	3	2	2	2	3	2
Justification for correlation	Apply basic Knowledge and fundamentals of data science for statistical description of data	The students will be able to identify the need for analysis of complex engineering problems	Able to design the process flow for developing data science applications.	The students able to recognize the need for statistics in the data science process	The students able to exhibit the knowledge in statistics	The students can Exhibit knowledge of data science in the field of IoT, Cloud Computing to develop IT solutions	The students able to exhibit knowledge in Standard Engineering practices.

- Images / Screenshot of the practice:



Figure 1: Reflection



Figure 2: Reflection

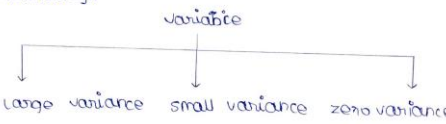
U. Thanishka Shree
B-Tech IT

08/10/23 Basic Statistical description of data

Statistical description of data:
Under this topic we learn about various topics

- 1) descriptive statistics
- 2) variability
- 3) correlation
- 4) probability distribution
- 5) normal distribution

1) descriptive statistics:
we gained knowledge about mean, median, mode for grouped and ungrouped data.

2) variability:


3) correlation:
this topic explains the relationship between two variables.

4) probability distribution:
it is a likelihood of all possible events, with two types as
1) Independent

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Basic Statistical Description of Data.

I am under this topic name is

- 1) Descriptive Statistics
- 2) Variability
- 3) Correlation
- 4) Probability Distribution
- 5) Normal Distribution

Overview of common errors
*I learnt the basic errors how to handle the errors in error description.

And different units of measurement different levels of Aggregation.

And also under the topic Three-tier house Architecture and types of Data Architecture.

Figure 3: Student Response Ms. U. Thanishka Shree and Swathi M

• **Reflective Critique:**

❖ **Feedback of practice from students and other stakeholders:**

- Students said that they were able to recollect and remember the concepts of statistics and the importance of statistics data science
- The students said that the activity for help to recollect the concepts that are taught in the previous class, the activity helps for improving self learning skills.
- It makes the students to gain knowledge on statistics.

❖ **Benefit of the practice:**

- Students can able to recall the formulas for finding Mean, Median, Mode, Standard Deviation and it improves the problem-solving skills.
- The students can share their doubts with neighbors.

❖ **Challenges faced in implementation:**

- Some questions were difficult for students to understand, and many struggled to formulate and articulate their doubts in clear English sentences.
- I felt the slow learners were not understood clearly. I explained separately for that students, then they understood clearly.

References:

- ❖ https://www.ritrjpm.ac.in/images/computer-science/2022-2023/1.Reflection_DBMS_KVS.pdf
- ❖ <https://ohioline.osu.edu/>
- ❖ <https://www.ritrjpm.ac.in/images/computer-science/>

Signature of Faculty Member**HOD**